

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - 1 - basic properties

- From Ohmi *et al*, Phys. Rev. A 92 (2015), we get

$$\pi_i^s(r) = \frac{1}{\sqrt{\omega_i}} \prod_{\substack{k=1 \\ k \neq i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \quad \pi_i^s(r_k^s) = \delta_{ik} / \sqrt{\omega_i^s}, \quad (1)$$

$$\left. \frac{d\pi_i^s(r)}{dr} \right|_{r=r_j^s} = \frac{1}{\sqrt{\omega_i}} \begin{cases} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq i, j}}^{N_p} \frac{r_j^s - r_k^s}{r_i^s - r_k^s}, & i \neq j, \\ \frac{1}{2} \left(\left[\pi_i^s(r_{N_p}^s) \right]^2 - \left[\pi_i^s(r_1^s) \right]^2 \right), & i = j, \end{cases} \quad (2)$$

- N_s sectors $r_0 \leq \dots r_s \leq r_{N_s}$ and N_p nodes a sector, which makes the number of point N so

$$N = N_s(N_p - 1) + 1 \quad (3)$$

$$n = (s - 1)(N_p - 1) + i \quad (4)$$

with

$$i = 1, \dots, N_p, \quad s = 1, \dots, N_s, \quad n = 1, \dots, N$$

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The FEDVR basis - 2 - The global basis

- The global FEDVR basis is defined, for $n = (s - 1)(N_p - 1) + N_p$, with $s = 1, \dots, N_s - 1$

$$\Pi_n(r) = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} \pi_{N_p}^s(r), & r \in [r_{s-1}, r_s], \\ \sqrt{\omega_1^{s+1}} \pi_1^{s+1}(r), & r \in [r_s, r_{s+1}], \end{cases} \quad \Omega_n = \omega_{N_p}^s + \omega_1^{s+1}, \quad (5)$$

This function bridges the sectors s and $s + 1$ and is continuous through the boundary between them (but its derivative is not).

- For the other values of n corresponding to the quadrature points inside sectors, i.e.

$$n = (s - 1)(N_p - 1) + i \quad (6)$$

and at the outer boundary $r_N = r_m$ of the box, we have

$$\Pi_n(r) = \pi_i^s(r), \quad r \in [r_s, r_{s+1}], \quad \Omega_n = \omega_i^s, \quad (7)$$

$$\Pi_n(r_{n'}) = \delta_{nn'} / \sqrt{\Omega_n}, \quad (8)$$

$$\int_0^{r_m} \Pi_n(r) f(r) \Pi_{n'}(r) dr \sim f(r_n) \delta_{nn'}. \quad (9)$$

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The FEDVR basis - 3 - the derivative

- We can extend Eq. (2) to any r of the domain and write

$$\frac{d\pi_i^s(r)}{dr} = \frac{1}{\sqrt{\omega_i^s}} \sum_{\substack{j=1 \\ j \neq i}}^{N_p} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq j, i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \quad (10)$$

- In terms of Π_n , the derivative is expressed as, for $n = (s-1)(N_p-1) + N_p$, with $\Omega_n = \omega_{N_p}^s + \omega_1^{s+1}$

$$\frac{d\Pi_n(r)}{dr} = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} d\pi_{N_p}^s(r)/dr, & r \leq r_s, \\ \sqrt{\omega_1^{s+1}} d\pi_1^{s+1}(r)/dr, & r \geq r_s, \end{cases} \quad (11)$$

and reduced to (10) otherwise.

The quantities we are looking for are, with the proper normalization

$$\psi(x, t) = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \Pi_n(x), \quad \text{and} \quad \frac{d}{dx} \psi(x, t) = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \frac{d}{dx} \Pi_n(x). \quad (12)$$

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The FEDVR basis - The nodes

$$N_s = 4, N_p = 5, \quad N = 15, \quad N' = 17.$$

$$n' = (s - 1)(N_p - 1) + i$$

$$N = N_s(N_p - 1) - 1, \quad N = 15$$

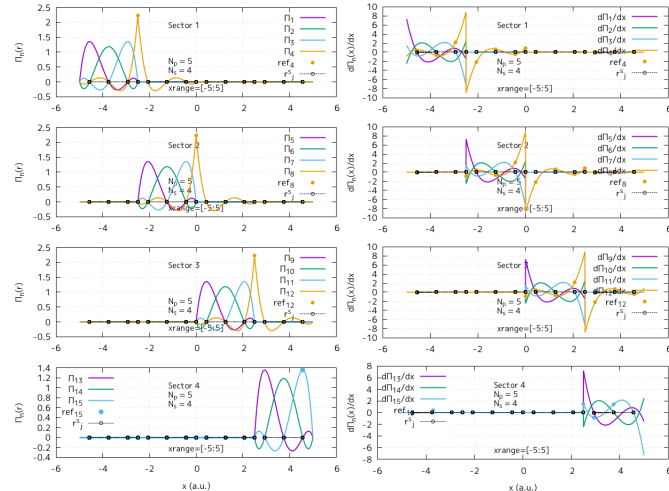
$$N' = N_s(N_p - 1) + 1 \quad N' = 17.$$

- $\mathcal{B}'$ provides mathematical support
- $\mathcal{B}$ is where we have the expression of the wavefunction
- We note that $n' = n + 1$ when defined

$\mathcal{B}'$				$\mathcal{B}$			
n'	s'	i'	$x_{n'}$	n	s	i	x_n
1	1	1	-5.0000000000000000	1	1	1	-4.5683170883849709
2	1	2	-4.5683170883849709	2	1	2	-3.7500000000000000
3	1	3	-3.7500000000000000	3	1	3	-2.9316829116150291
4	1	4	-2.9316829116150291	4	1	4	-2.5000000000000000
5	1	5	-2.5000000000000000				
5	2	1	-2.5000000000000000	5	2	1	-2.0683170883849713
6	2	2	-2.0683170883849713	6	2	2	-1.2500000000000000
7	2	3	-1.2500000000000000	7	2	3	-0.43168291161502864
8	2	4	-0.43168291161502864	8	2	4	0.0000000000000000
9	2	5	0.0000000000000000				
9	3	1	0.0000000000000000	9	3	1	0.43168291161502864
10	3	2	0.43168291161502864	10	3	2	1.2500000000000000
11	3	3	1.2500000000000000	11	3	3	2.0683170883849713
12	3	4	2.0683170883849713	12	3	4	2.5000000000000000
13	3	5	2.5000000000000000				
13	4	1	2.5000000000000000	13	4	1	2.9316829116150291
14	4	2	2.9316829116150291	14	4	2	3.7500000000000000
15	4	3	3.7500000000000000	15	4	3	4.5683170883849709
16	4	4	4.5683170883849709				
17	4	5	5.0000000000000000				

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The FEDVR basis - Π_n and its derivative

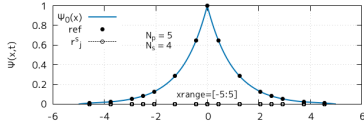


$$\Pi_n(x) = \sum_{m=1}^N \Pi_n(x_m) \hat{e}_m \quad (13)$$

- The continuous line represent Eq. (5) for an arbitrary x in the sector's range
- The colored dot represents the value for the different nodes x_j , i.e. Eq. (13), with $\hat{e}_m$ the unitary vectors of the cartesian basis

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The FEDVR basis - The analytical wavefunction



$$\pi_i^s(r) = \frac{1}{\sqrt{\omega_i}} \prod_{\substack{k=1 \\ k \neq i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s},$$

$$\pi_i^s(r_k^s) = \delta_{ik} / \sqrt{\omega_i^s},$$

$$\Pi_n(r) = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} \pi_{N_p}^s(r), & r \in [r_{s-1}, r_s], \\ \sqrt{\omega_1^{s+1}} \pi_1^{s+1}(r), & r \in [r_s, r_{s+1}], \end{cases} \quad \Pi_n(x_m) = \delta_{nm} / \sqrt{\Omega_n},$$

for $n = (s-1)(N_p-1) + N_p$ and reduces to $\pi_i^s(r)$ otherwise

- The black dots represent (14) for $m = 1, \dots, N$
- The blue line represent (15) in the whole range

$$\psi(x_m, t) = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \Pi_n(x_m) = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \delta_{nm} / \sqrt{\Omega_n} \quad (14)$$

$$\psi(x, t) = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \Pi_n(x) \quad (15)$$

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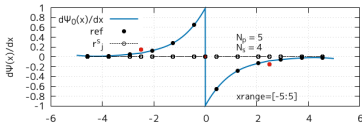
The FEDVR basis - The analytical derivative of the wavefunction

$$\left. \frac{d\pi_i^s(r)}{dr} \right|_{r=r_j^s} = \frac{1}{\sqrt{\omega_i}} \begin{cases} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq i,j}}^{N_p} \frac{r_j^s - r_k^s}{r_i^s - r_k^s}, & i \neq j, \\ \frac{1}{2} \left(\left[\pi_i^s(r_{N_p}^s) \right]^2 - \left[\pi_i^s(r_1^s) \right]^2 \right), & i = j, \end{cases}$$

$$\frac{d\Pi_n(r)}{dr} = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s} d\pi_{N_p}^s(r)/dr, & r \leq r_s, \\ \sqrt{\omega_1^{s+1}} d\pi_1^{s+1}(r)/dr, & r \geq r_s, \end{cases} \quad \text{with} \quad \frac{d\pi_i^s(r)}{dr} = \frac{1}{\sqrt{\omega_i^s}} \sum_{\substack{j=1 \\ j \neq i}}^{N_p} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k \neq j,i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s},$$

for $n = (s-1)(N_p-1) + N_p$ and reduces to $d\pi_i^s(r)/dr$ otherwise

- The black dots represent (16) for $m = 1, \dots, N$ (ref)
- The blue line represents (17) in the whole range



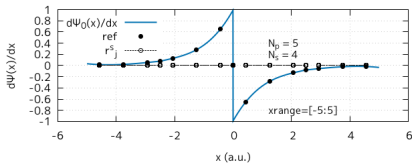
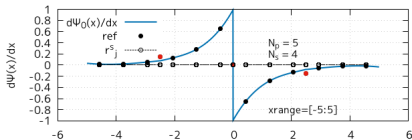
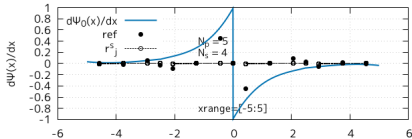
The values for $n' = (s-1)(N_p-1) + N_p$ are off

$$\frac{d\psi(x_m, t)}{dx} = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \frac{d\Pi_n(x_m)}{dx} \quad (16)$$

$$\frac{d\psi(x, t)}{dx} = \sum_{n=1}^N \sqrt{\omega_i^s} c_n(t) \frac{d\Pi_n(x)}{dx} \quad (17)$$

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The FEDVR basis - Extension by continuity



- (Up) What was shown last week
- (Middle) Improvement of the discrete formula
- (Down) Fixing the values at the boundaries

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Propagators

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (18)$$

- So we want our drivers to solve the following general case as efficiently as possible

$$y' = Ay + S(y, t), \quad (19)$$

$$(20)$$

- We are going to build a couple of them and rigorously assess their convergence

1. Runge-Kutta 4 (IFRK4)
2. Split-operator
3. Adams methods
4. etc...

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Convergence criteria



$$y' = Ay + S(y, t). \quad (21)$$

- We assess our propagation drivers based on two different tests

- Absolute convergence

$$E_\ell^{\text{abs}} = |y_\ell(T) - y_{\text{exact}}(T)|, \quad (22)$$

$$p_\ell^{\text{abs}} = \frac{\log E_{\ell-1}^{\text{abs}} / E_\ell^{\text{abs}}}{\log(2)} \quad (23)$$

- Self-convergence

$$D_\ell = |y_\ell(T) - y_{\ell-1}(T)|, \quad D_\ell \sim Ch_\ell^p \quad (24)$$

$$p_\ell^{\text{self}} = \frac{\log(D_{\ell-1}/D_\ell)}{\log(2)} \quad (25)$$

where the number of steps N , and the time step h are such as

$$N_\ell = N_0 2^\ell, \quad h_\ell = \frac{T}{N_0 2^\ell}. \quad (26)$$

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Runge Kutta

Level	Euler	RK2	RK4
3	~ 1.0	~ 2.2	~ 4.0
4	~ 1.0	~ 2.1	~ 4.0

- The above table shows the self-convergence data p_ℓ^{self} for the following forced Riccati equation

$$y' = \lambda y + \alpha y^2 + \beta e^{\omega t} \quad (27)$$

- The absolute convergence table, pertaining to p_ℓ^{abs} , shows the same data as long as $\beta = 0$
- The same table will be dressed for at least 3 production ready drivers
- "Ready for production" means ready to be used in the main program